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CBCS SCHEME

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**Sixth Semester B.E. Degree Examination, July/August 2022**

# Sustainability Concepts in Civil Engineering

Time: 3 hrs. Max. Marks: 100

*Note: Answer any FIVE full questions, choosing ONE full question from each module.*

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

## Module-1

1 a. Define sustainability and discuss the need for sustainable developments. **(06 Marks)**
b. Explain:
(i) Water Control and Prevention Act
(ii) Air Control and Prevention Act **(14 Marks)**

**OR**

2 a. Explain the three-pillar model of sustainability. **(08 Marks)**
b. What are the challenges to attain sustainable developments? **(06 Marks)**
c. What is MEA? List the key features. **(06 Marks)**

## Module-2

3 a. Explain briefly the concept of Local Environmental issues. **(10 Marks)**
b. Write short notes on:
(i) Carbon foot print
(ii) Carbon Trading **(10 Marks)**

**OR**

4 a. Write a short note on Life Cycle Analysis. **(10 Marks)**
b. Explain:
(i) Global Warming
(ii) Ozone Layer Depletion **(10 Marks)**

## Module-3

5 a. Define Green Building. Discuss any two certification programmes implemented in India for Green Building. **(12 Marks)**
b. Write a short note on Passive Solar Building Design. **(08 Marks)**

**OR**

6 a. How energy efficiency can be achieved in a sustainable building design? Explain. **(10 Marks)**
b. Explain briefly:
(i) Thermal storage **(05 Marks)**
(ii) Sustainable cities and sustainable transport **(05 Marks)**

## Module-4

7 a. Define Energy. Explain various forms of energy. **(10 Marks)**
b. Write short notes on:
(i) Geothermal Energy **(05 Marks)**
(ii) Wind Energy **(05 Marks)**



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**OR**

8 a. Briefly explain the methods of extraction of energy from oceans. **(10 Marks)**
b. Solar energy as a renewable form of energy. Explain briefly. **(10 Marks)**

## Module-5

9 a. Write a short note on sustainable urbanization. **(10 Marks)**
b. Explain briefly the concept of sustainable Poverty Reduction. **(10 Marks)**

**OR**

10 a. Briefly explain the Social and Technological change for Sustainable Development. **(10 Marks)**
b. Write short notes on:
(i) Industrial Ecology **(05 Marks)**
(ii) Industrial Symbiosis **(05 Marks)**

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